

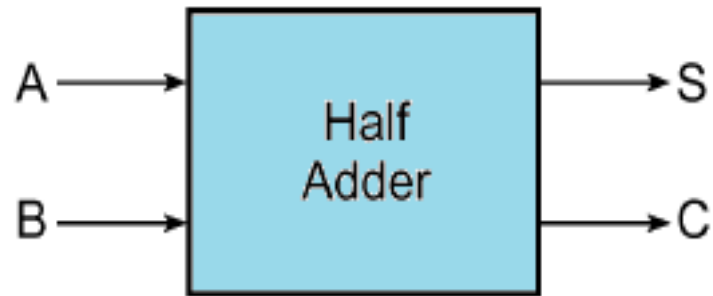
2.3 Arithmetic circuits

Half Adder

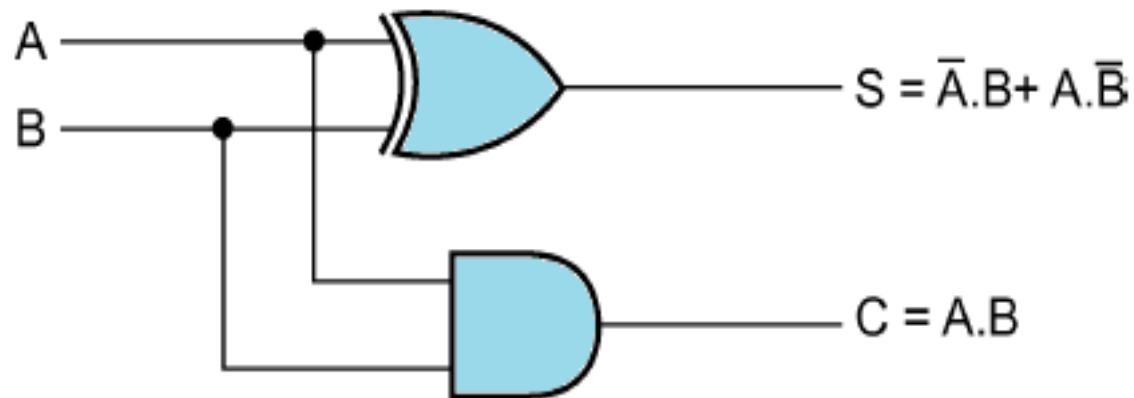
- Half Adder-A logic circuit for addition of two one-bit numbers
- Truth table

Inputs		Outputs	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

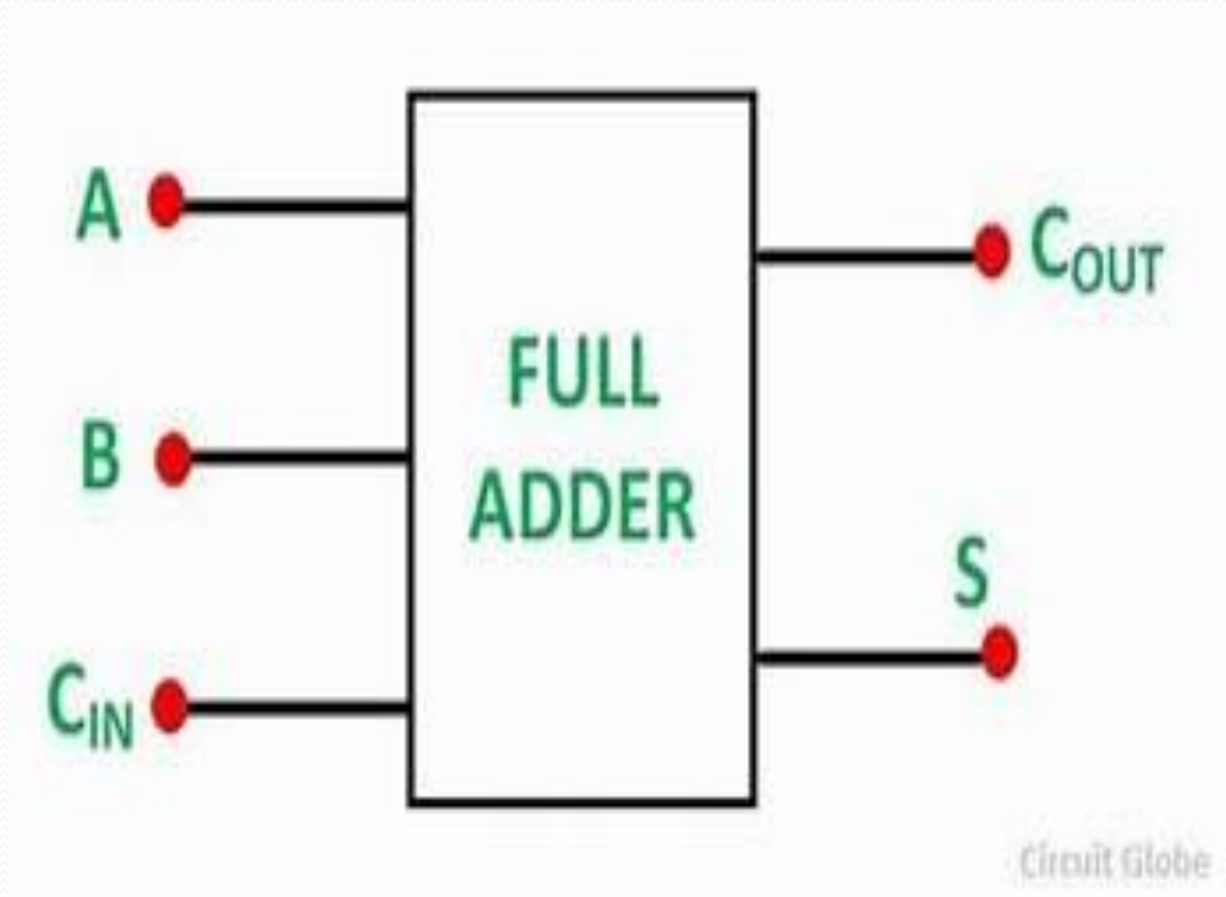
- $\text{Sum} = A'B + AB' = A \text{ ex-or } B$
- $\text{Carry} = A.B$



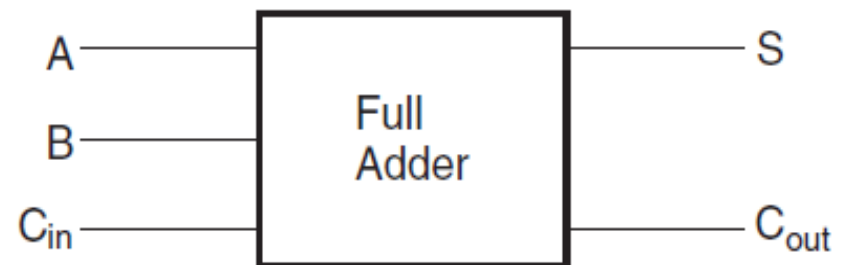
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



Schematic of full adder



A	B	C_{in}	SUM (S)	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



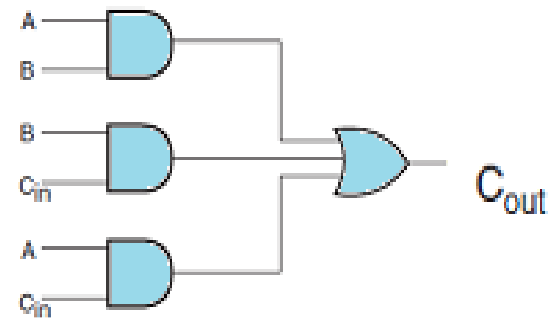
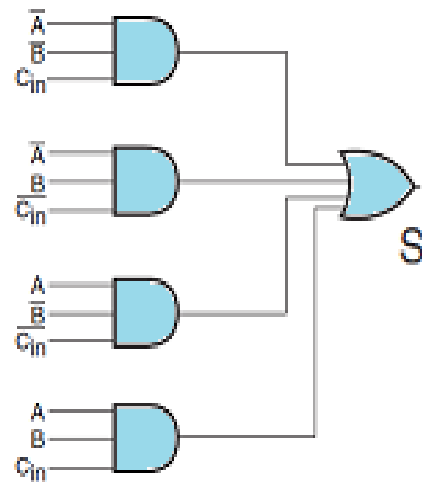
AB \ C_{in}	$\overline{C_{in}}$	C_{in}
$\overline{A}\overline{B}$		
$\overline{A}B$		1
AB	1	1
$A\overline{B}$		1

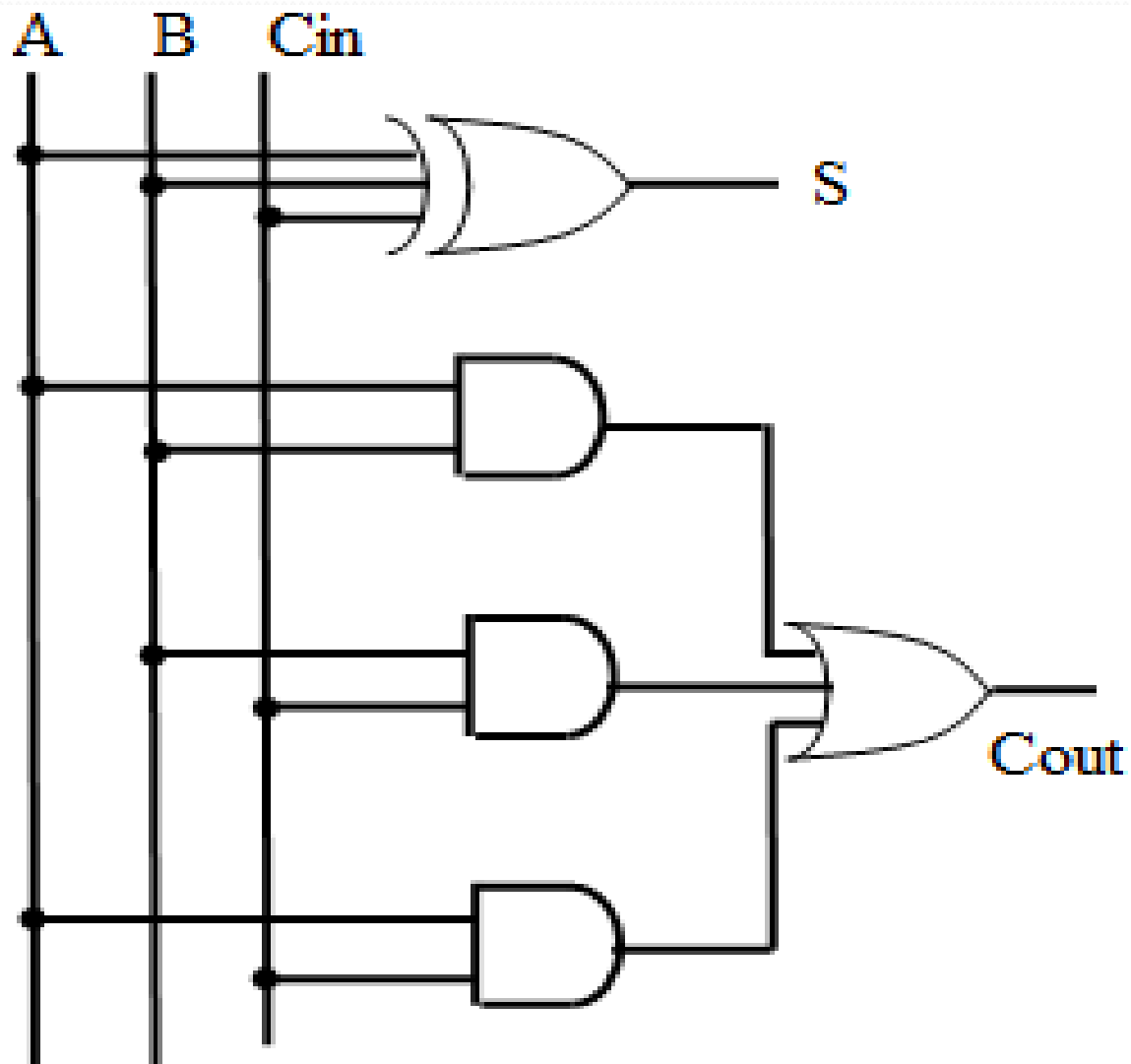
AB \ C_{in}	$\overline{C_{in}}$	C_{in}
$\overline{A}\overline{B}$		1
$\overline{A}B$	1	
AB		1
$A\overline{B}$	1	

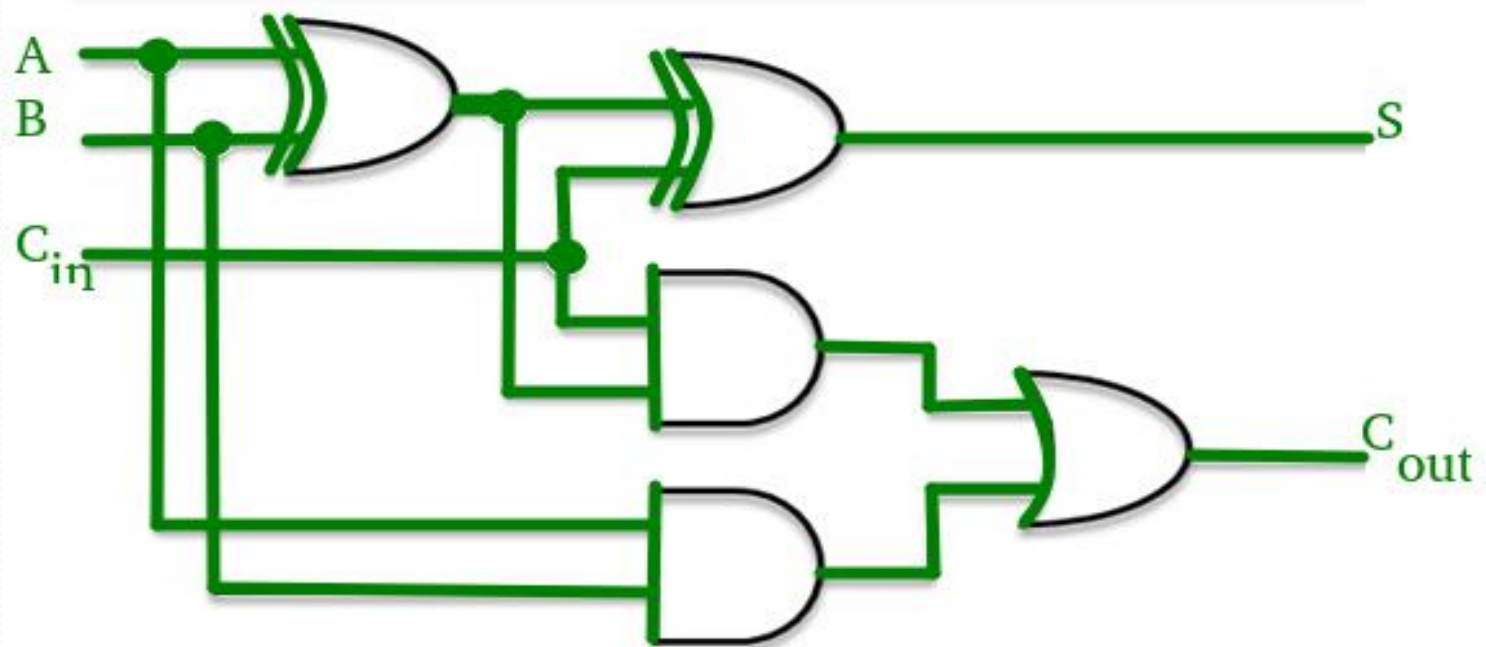
$$C_{out} = \overline{A}.B.C_{in} + A.\overline{B}.C_{in} + A.B.\overline{C_{in}} + A.B.C_{in} \quad S = \overline{A}.\overline{B}.C_{in} + \overline{A}.B.\overline{C_{in}} + A.\overline{B}.\overline{C_{in}} + A.B.C_{in}$$



A	B	C_{in}	SUM (S)	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



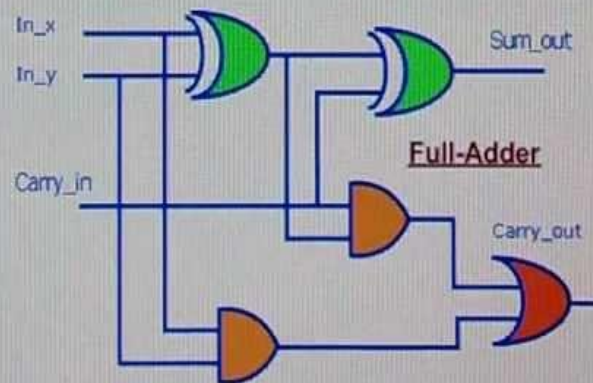
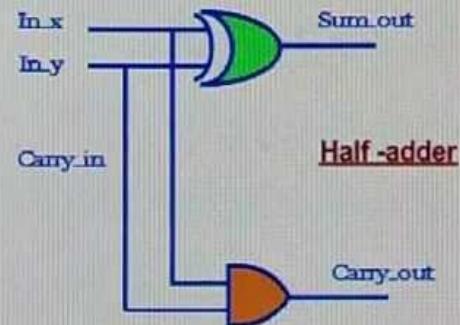




Full Adder Truth Table and circuit

The circuit involves
2 half-adders & one OR gate.
Alternately,
2 XOR gates, 2 AND gates
and 1 OR gate.

In_x	In_y	Carry_in	sum_out	Carry_out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



USES OF FULL ADDER

- Basic building block of on-chip libraries.
- Configured according to desired complexity of arithmetic and numeric computations.
- In processors and other kinds of computing devices, adders are used in the arithmetic logic unit.
- Adders also use in other parts of the processor, where they are used to calculate addresses, table indices, and similar operations.

Four bit adder

- $Y = A + B$

Cin
↓

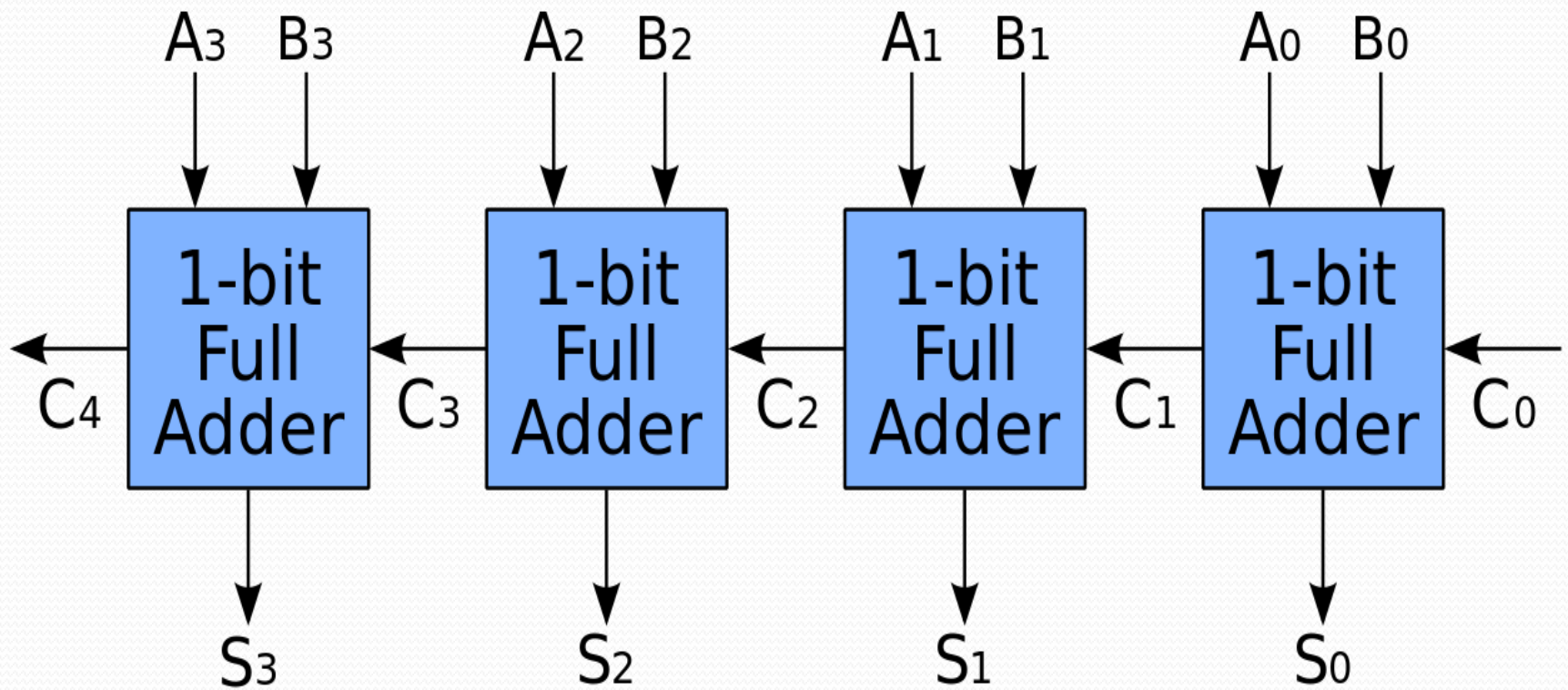
A₃ A₂ A₁ A₀

+ B₃ B₂ B₁ B₀

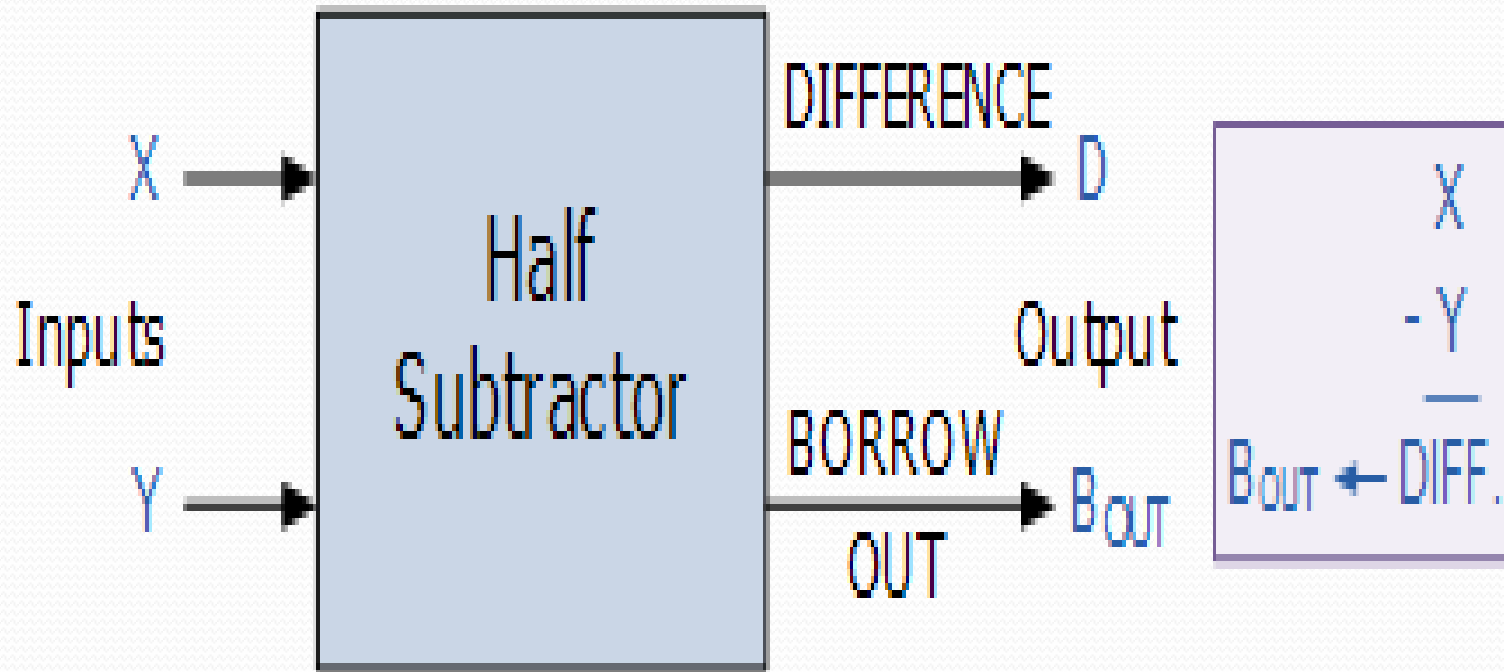
Cout ← S₃ S₂ S₁ S₀

- To implement this 4 bit addition, 4 Full adders are required
- 4 Full adders are cascaded.

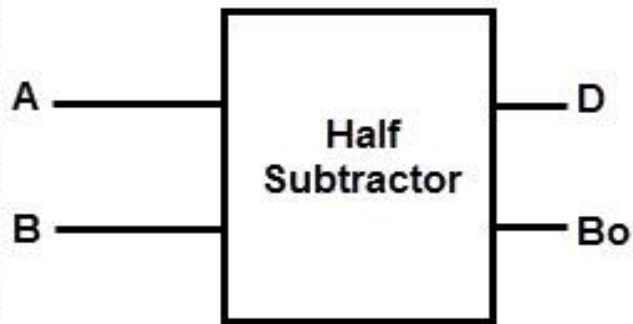
Schematic of 4 bit Full adder



Binary Subtractor



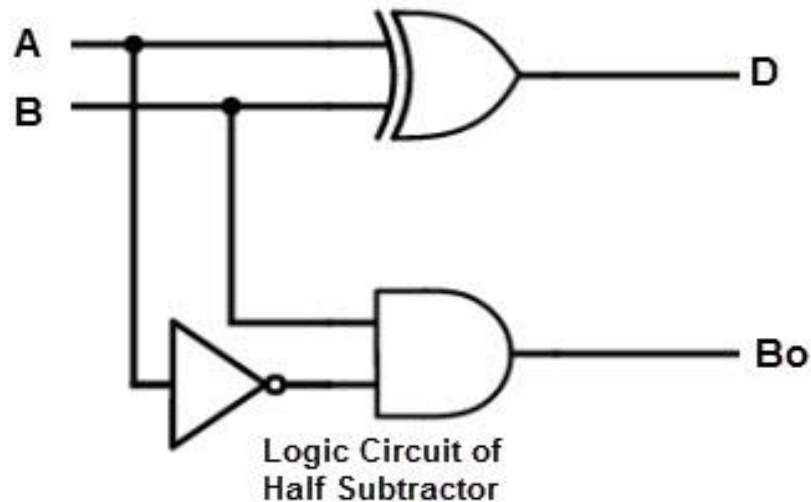
Half subtractor truth table and circuit



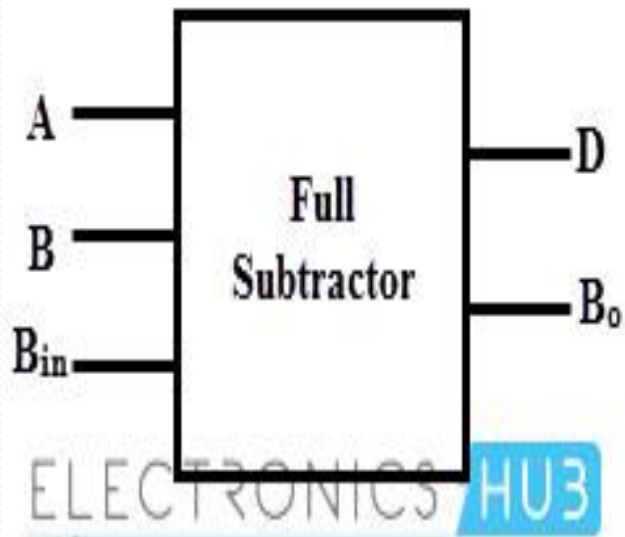
Block Diagram

A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Truth Table



Full subtractor

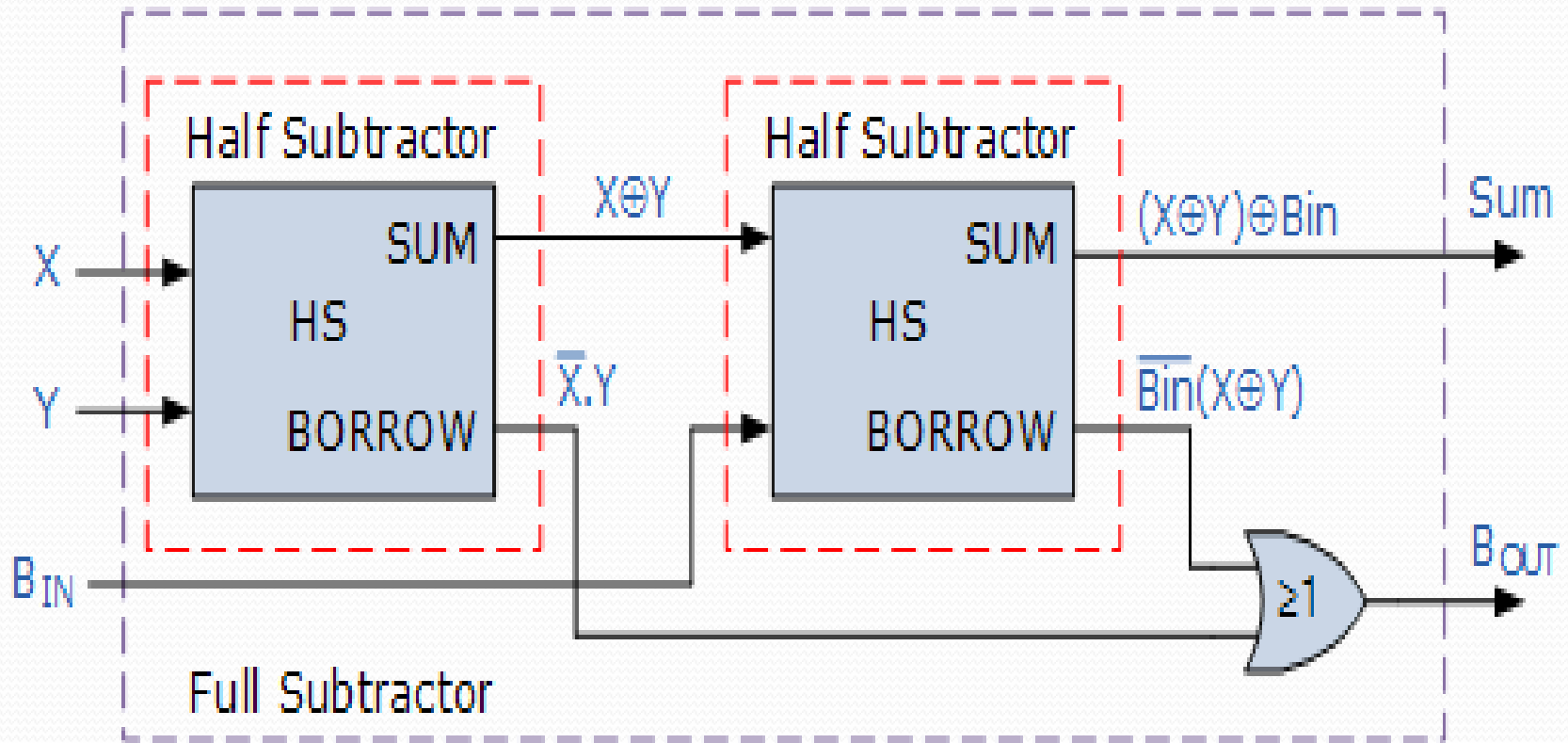


Block Diagram of
Full Subtractor

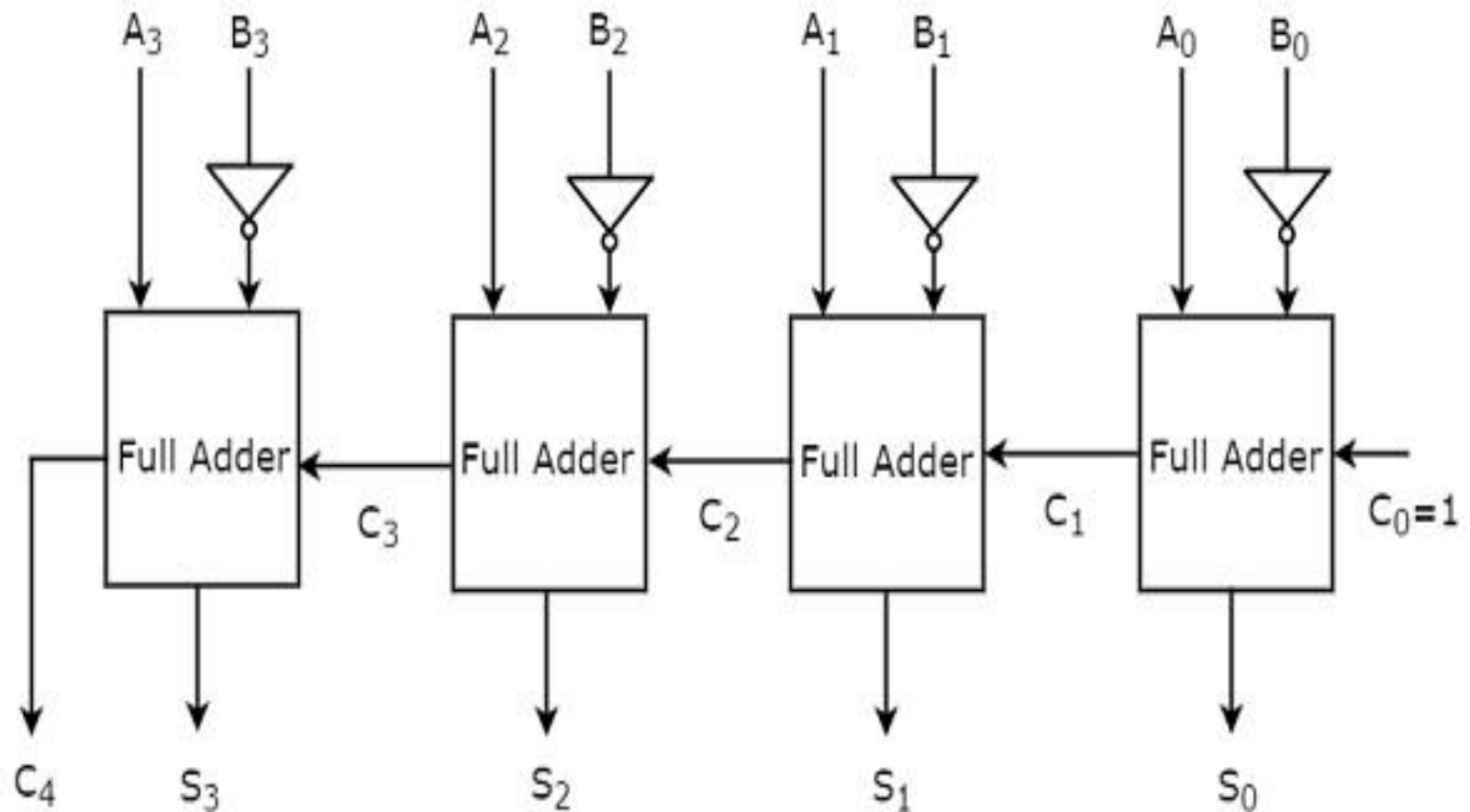
A	B	B_{in}	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

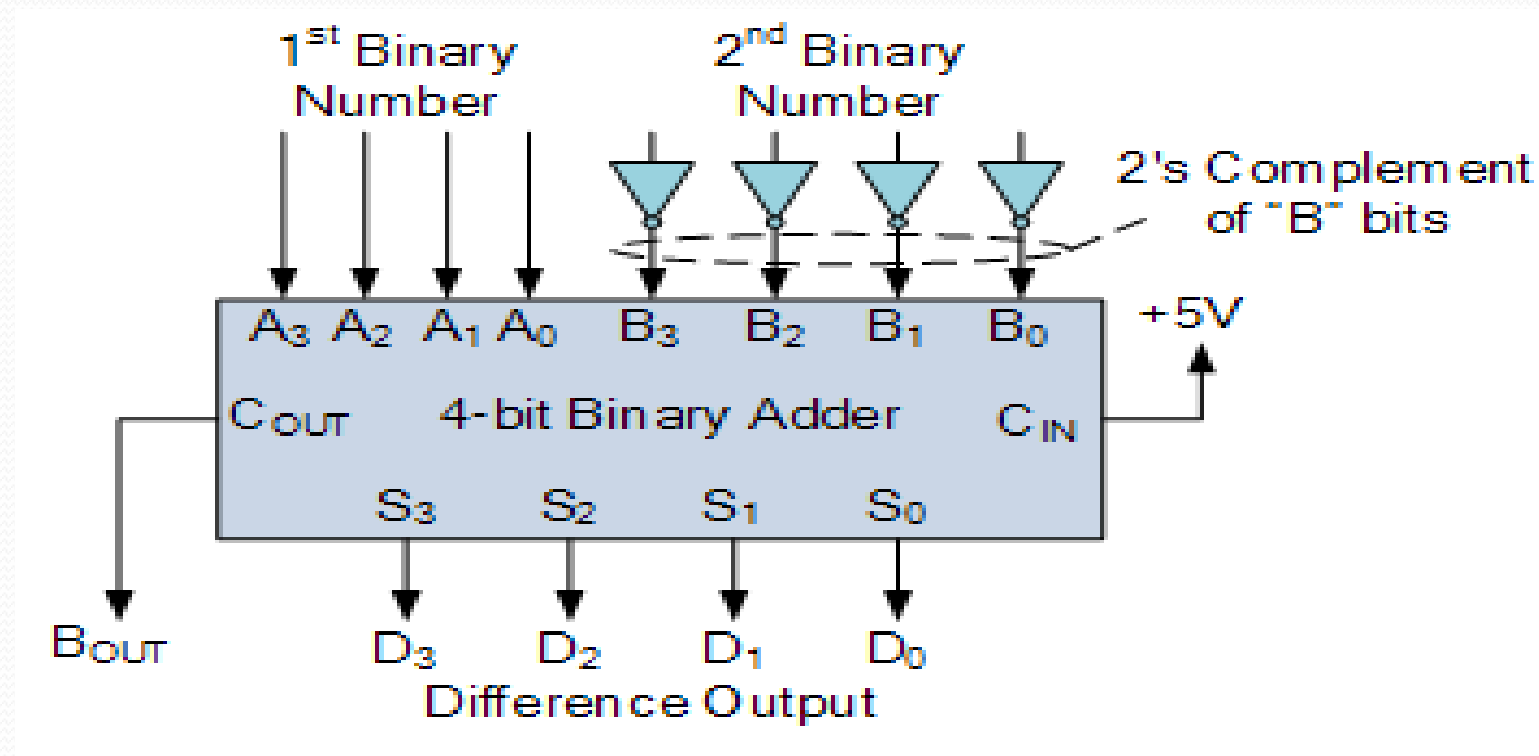
Truth Table

Full subtractor using half subtractor



Four bit subtractor using four bit full adder





Four bit adder-subtractor

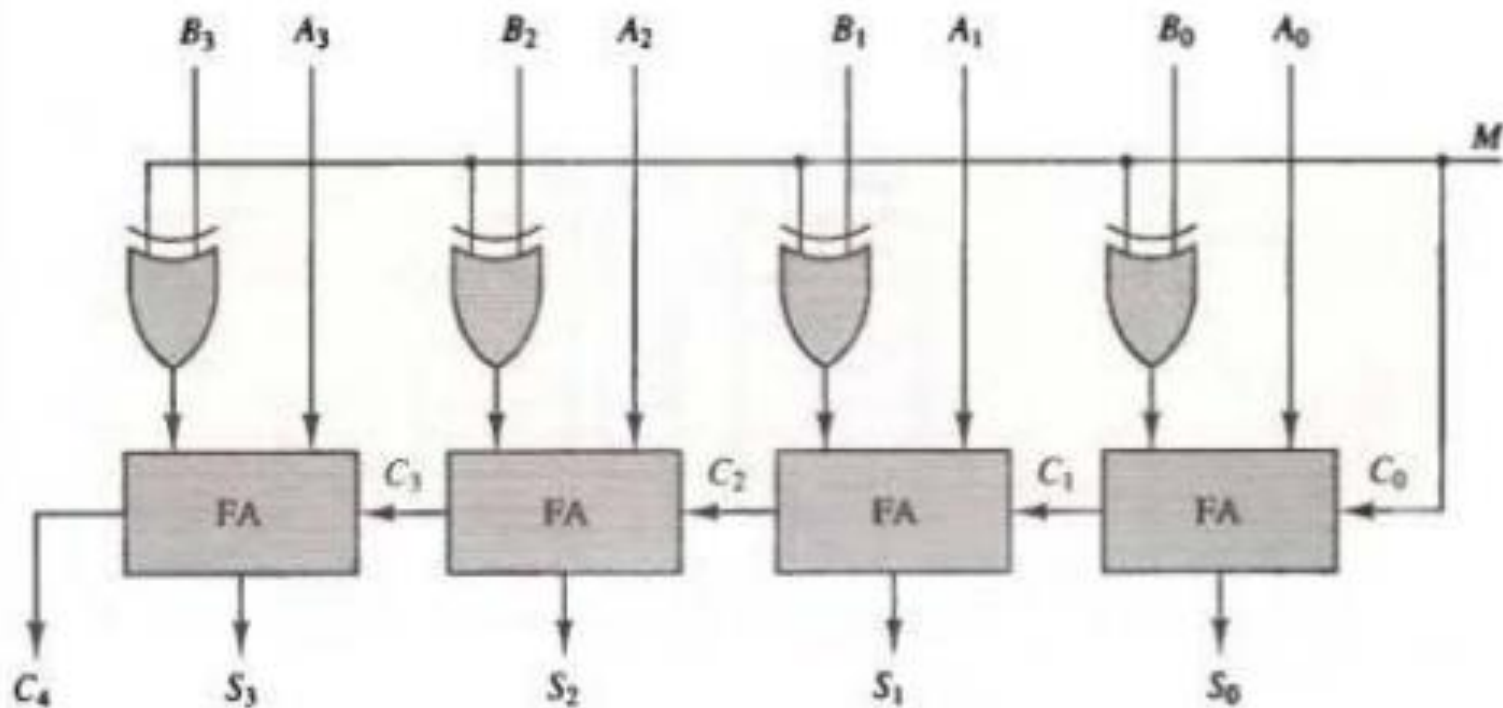


Figure 4-7 4-bit adder-subtractor.

DEFINITION



BCD adder A 4-bit binary **adder** that is capable of adding two 4-bit words having a **BCD** (binary-coded decimal) format. The result of the addition is a **BCD**-format 4-bit output word, representing the decimal sum of the addend and augends, and a carry that is generated if this sum exceeds a decimal value of 9.

BCD Adder

What is BCD ADDER?

- ✓ Also Known as 8421 digit.
- A 4-bit binary adder.
- Adds two 4-bit digits having a BCD.
- Resulting format 4-bit output digit.
- Sum exceeding decimal value of 9, a carry's generated.

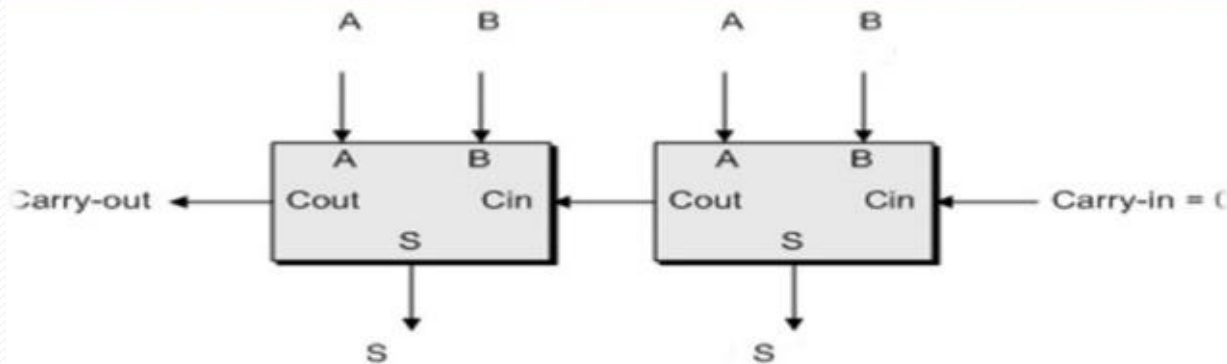


Figure 2: A 1-byte (2-digit) BCD adder.

WHY ADD 6 TO THE RESULT?

- Sum of two BCD digits exceeding 9.
- A carry is generated.
- Converting the invalid digit into valid digit.
- Carry generated by adding 6 to the invalid BCD digit's passed on to the next BCD digit.

```
  0101
+ 0110
-----
  1011 → Invalid BCD number
+ 0110 → Add 6
-----
0001 0001 → Valid BCD number
```

$$\begin{array}{rcl}
 \text{Carry from} & \rightarrow & 0 \\
 \text{previous digit} & & 7 \\
 & + & \underline{6} \\
 & & 13
 \end{array}$$

$$\begin{array}{r}
 0 \\
 0111 \\
 + \underline{0110} \\
 1101
 \end{array}$$

$$\begin{array}{rcl}
 \text{Carry to next digit} & \rightarrow & 1 \quad \begin{array}{r} 110 \\ \underline{0011} \end{array} \leftarrow \text{Add 6 because 1101 is not a valid BCD digit}
 \end{array}$$

$$\begin{array}{rcl}
 \text{Carry from} & \rightarrow & 1 \\
 \text{previous digit} & & 9 \\
 & & \underline{9} \\
 & & 19
 \end{array}$$

$$\begin{array}{r}
 1 \\
 1001 \\
 + \underline{1001} \\
 1 \quad 0011
 \end{array}$$

$$\begin{array}{rcl}
 \text{Carry to next digit} & \rightarrow & 1 \quad \begin{array}{r} 110 \\ \underline{1001} \end{array} \leftarrow \text{Add 6 because of carry to next digit}
 \end{array}$$

Examples of BCD addition

Non-BCD	$0101 = 5$
BCD correction	$+ \underline{0111} = +7$
In BCD	$1100 = 12$
	$+ \underline{0110} = +6$
	$1\ 0010 = 12$

Non-BCD	$1001 = 9$
BCD correction	$+ \underline{0110} = +6$
In BCD	$1111 = 15$
	$+ \underline{0110} = +6$
	$1\ 0101 = 15$

Non-BCD	$1001 = 9$
BCD correction	$+ \underline{1001} = +9$
In BCD	$1\ 0010 = 18$
	$+ \underline{0110} = +6$
	$1\ 1000 = 18$

Correct: Result
is in BCD.

$$\begin{array}{r} 0110 = 6 \\ + \underline{0011} = +\underline{3} \\ \underline{1001} = \underline{9} \end{array}$$

Wrong: Result is
not in BCD.

$$\begin{array}{r} 0101 = 5 \\ + \underline{0111} = +\underline{7} \\ \underline{1100} = \underline{12} \end{array}$$

BCD Adder schematic using binary adder

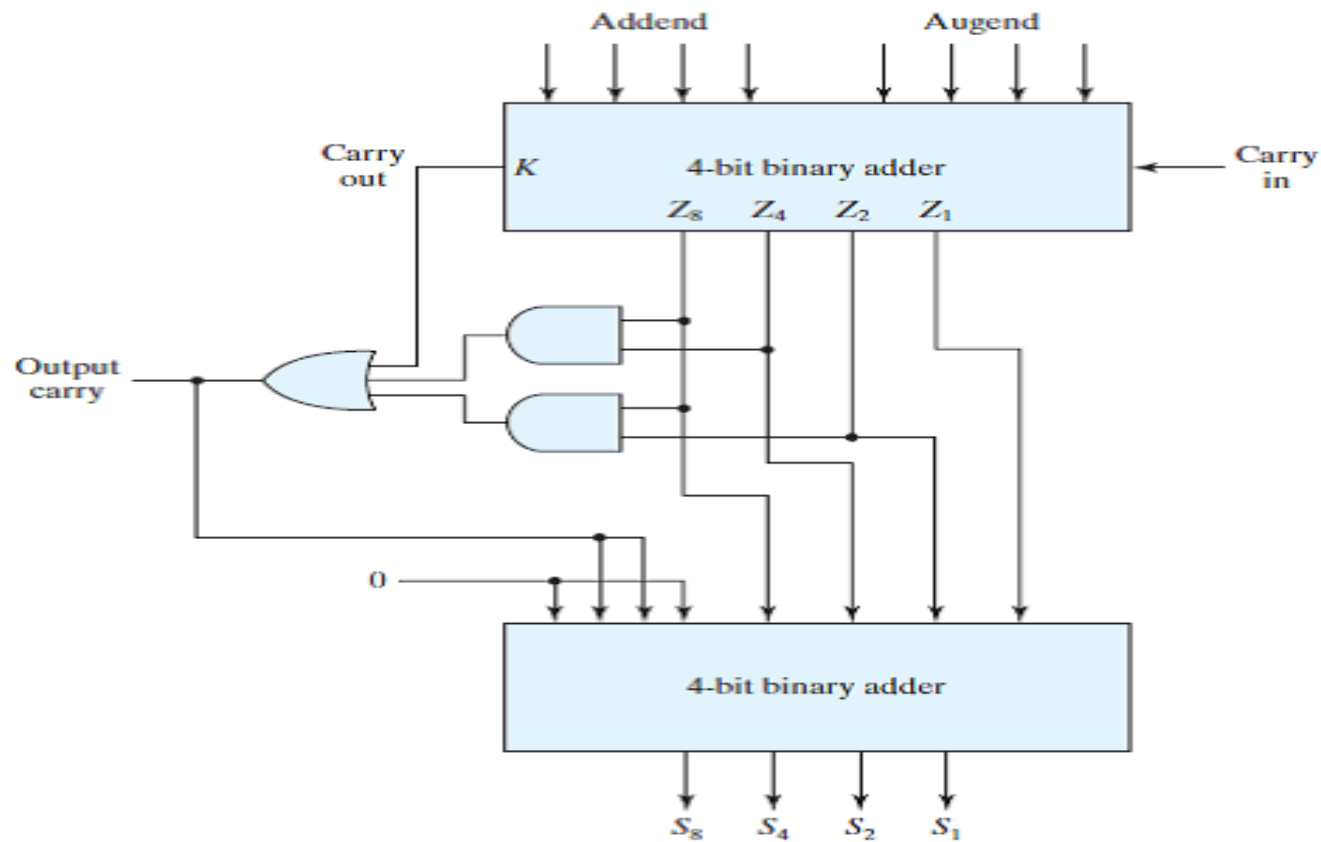
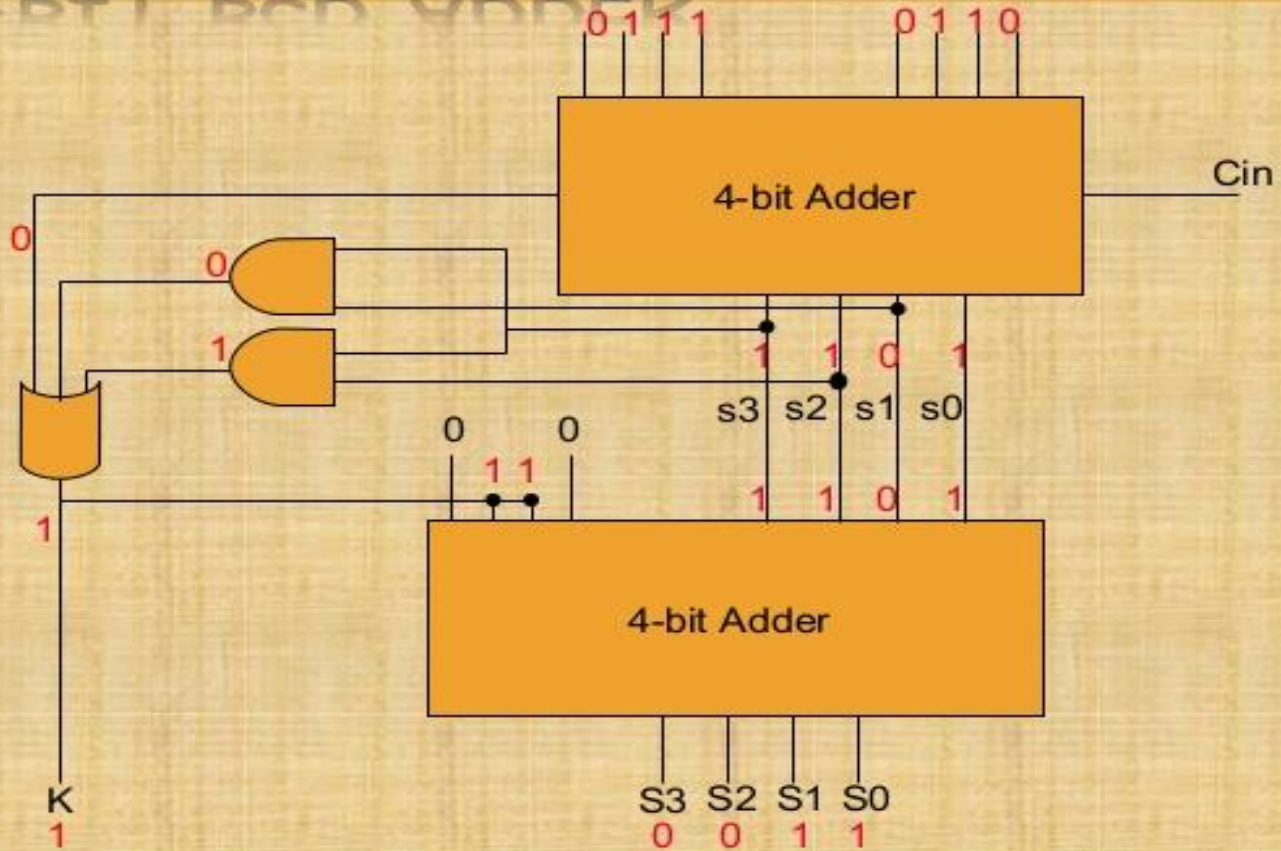


FIGURE 4.14
Block diagram of a BCD adder

4-BIT BCD ADDER



THANK YOU